

# Manroj Gill

780-257-9105 | [manrojjill@hotmail.com](mailto:manrojjill@hotmail.com) | [linkedin.com/in/manroj-gill](https://www.linkedin.com/in/manroj-gill) | [github.com/M-GII](https://github.com/M-GII)

## Education

### University of Alberta

Computer Engineering, BSc Co-op

Edmonton, Alberta

Sept 2023 — April 2028

### Relevant Coursework:

- **Programming:** Object-Oriented Software Design, Introduction to File and Database Management, Introduction to Tangible Computing I & II, Operating System Concepts
- **Hardware:** Electronic Devices, Computer Organization and Architecture, Introduction to Microprocessors, Electrical Circuits I & II

## Projects

### VoltCart | React, Node.js, Express, MongoDB, Tailwind CSS

- Developed a full-stack e-commerce platform with a React frontend and RESTful Node.js/Express backend
- Designed and implemented backend APIs with MongoDB/Mongoose for product, user, and order management, ensuring efficient data handling and persistence
- Designed and implemented JWT-based authentication with protected routes to secure admin access and operations.
- Developed an admin dashboard to manage product inventory and monitor customer orders
- Integrated Cloudinary for image uploads and storage

### Quizzical | React, Node.js, Express, MongoDB, Open Trivia Database API

- Developed a full-stack quiz application with dynamic question generation and persistent user score tracking
- Built a Node.js/Express backend with JWT authentication for secure user data handling and session management across requests
- Designed APIs for score submission and retrieval of recent and top performance metrics, supporting leaderboard-style features
- Integrated Open Trivia Database API with dynamic parameters to generate customizable quizzes

### Assembly Endgame | React, JavaScript

- Designed a word-guessing game with 8 attempts, removing one programming language per incorrect guess until only Assembly remains.
- Added interactive features including unique loss messages for each language, conditional UI styling with clsx, confetti animations, a win state, and a “New Game” reset option to enhance user experience.
- Implemented structured game-state management using React hooks to control word selection, guessed letters, win/loss logic, and dynamic UI rendering.

## Experience

### Software Team Member, AlbertaLoop — Edmonton, Alberta

Sept 2025 — Dec 2025

- Contributed to software development for pod control by assisting with Finite State Machine (FSM) design and basic fault-handling logic.
- Used Python to complete small computer vision tasks related to track fault detection and perception
- Assisted with setup, testing, and debugging of Raspberry Pi-based systems used in software prototyping
- Collaborated with team members to understand system behavior, software structure, and integration challenges in a multi-component project.

### Document Control Assistant, 1717616 ALBERTA LTD — Edmonton, Alberta

Mar 2022 — Aug 2024

- Managed and organized high-volume records with a focus on accuracy, version control, and efficient information retrieval
- Coordinated cross-team documentation workflows and produced clear, reliable reports while handling sensitive data securely

## Skills

**Languages:** JavaScript, Python, Java, C/C++, SQL

**Frameworks & Libraries:** React.js, Node.js, Express.js, Tailwind CSS

**Developer & Technical Tools:** Visual Studio Code, GitHub, LTSpice, WaveForms, MATLAB, Excel, Word

**Certifications:** HTML/CSS, JavaScript, React, Node.js, Express.js